Aftershock

Building Safety Coordinator · Kestrel Bay

15 DAYS · 52 STOPS · GRADE 12 · AFTERSHOCK

THE OPENING CARD

Three days after a magnitude 6.8 earthquake, four hundred Kestrel Bay households are still sleeping in halls. Every building has a card on its door: green for use, yellow for restricted use, red for nobody inside. You are the building safety coordinator, and those colours are your decisions. Aftershocks are still coming. If a card is too confident, somebody can walk back into a building that cannot take the next shake. If it is too cautious, homes stay empty, a school stays shut and hospital care moves sixty kilometres away. In fifteen days you sign the Placard Register: what every building may be used for, what evidence supports that call, and what nobody has checked yet.

Two towns, one earthquake

BEFORE THE DAY — TRIAL

Walk Upper Town before you placard anything

Three days after, and you have not been here before. Okonkwo, the structural engineer, takes every assessor round the near ground once on foot: the office, the structural bench, the materials lab. Nothing is placarded from a map, and the plan card names streets that mean nothing until you have stood in them and seen which way the ground falls.

PLAN CARD 138W

Friday, three days after the quake. The first thing anybody says at the base is that none of this makes sense. Upper Town sits on a granite bench. It has parapets in the street and almost every building still standing. The Flats sit 1100 metres away, on the same side of the same fault. There is sand in the gutters there. A six-storey block leans at 8 degrees. A wharf has moved 2 metres. Marisol Okonkwo, the structural engineer, will not sign an assessment programme until she can say why. Elena Navarro, the geotechnical engineer, thinks she knows already. She wants a borehole to prove it. You have two records of the same shaking. Work out what the ground did to that shaking on its way up. Today you decide which half of town gets the assessors first.

BRIEFING 15W

One earthquake hit Kestrel Bay. The ground under the Flats made it a different disaster.

WHAT YOU DECIDE 27W

Explain why the same earthquake made the Flats far more dangerous than Upper Town, so the first reassessment effort goes where it can change who gets home.

1.1 What knocking it down would fix

PUBLIC SAFETY · A ROOM · VALUE

SCENE 46W

Halvorsen, the district emergency manager, wants Marina Court demolished this week. Delacroix, the ward councillor for the Flats, has 12 neighbouring buildings on the same reclaimed ground and 5 investigation-days before the cordon review. Four proposed uses of that window are written across the site map.

GUIDE 61W

Five investigation-days, four proposals, and a demolition somebody wants this week. Open each proposal to see what decision its result could change. Marina Court is already cordoned, so evidence about that one building changes nothing that is not already decided — the twelve neighbours on the same reclaimed ground are where the open decision is. Commit the window you would spend.

ASK 12W

Which investigation buys information that generalises beyond the one building already cordoned?

BEHIND THE BUTTON 148W

What is already decided and what is not. Marina Court is cordoned; nobody is going back in this week whatever a test says. The open question is whether the ground condition that damaged it requires action on the twelve buildings around it, and that is a question about the ground rather than about the building.

Why the most visible damage is the trap. A proposal to instrument the worst-damaged structure produces vivid results and answers a question that has been closed. It is the most defensible-sounding way to spend five days and learn nothing that changes a decision.

What generalising means here. A campaign of cone tests and boreholes across the Flats measures the fill itself, which every one of the thirteen buildings sits on. One result then bears on twelve open decisions instead of one closed one — the same money, an order of magnitude more use.

TAKES AS READ

a decision can treat a symptom or a cause

VERDICT 81W

Demolition can remove Marina Court as a visible hazard, but it does not reveal whether the same ground condition extends beneath nearby buildings. A corridor investigation can. CPTs, boreholes and groundwater checks test the ground shared by several sites while the existing cordon controls the known building hazard. Extra lean monitoring or a one-building underpinning concept answer narrower questions. Value of information therefore depends on the decision the evidence can change, not on how dramatic or precise the proposed work sounds.

TAKEAWAY 26W

The best next measurement is the one that can change the decision you still have to make, not the one that studies the most visible damage.

1.2 What the tested anchors actually carried

MATERIALS & TESTING · A PERSON · BALLPARK

SCENE 31W

Kirsten Sørensen has pulled two cast-in anchors from a spare panel in the yard. The design drawing says 40 kilonewtons each. The first came out at 24, the second at 27.

GUIDE 67W

Five numbers, and two of them belong to other questions: the roof span and the number of panels. One is what the drawing promised rather than what the wall got. Ask of each whether it describes what the anchors carried or what the wall asks of them. And note what two pulls cannot carry: a ratio is not a factor of safety, whatever it comes out at.

ASK 10W

Estimate the observed capacity-to-demand ratio for the two tested anchors.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a designed capacity is a claim to be checked

VERDICT 89W

The two pulls average 25.5 kN against an assessed demand of 31 kN, so the observed specimen-capacity-to-demand ratio is about 0.82. That is enough to show that the tested connection detail cannot simply be accepted from the 40 kN drawing value. It is not a code resistance factor or a formal factor of safety: two specimens do not characterize variability, failure mode, deterioration or the required design margins. The operational decision does not need those questions settled today—keep the gym restricted, expand the testing, and design the connection repair.

TAKEAWAY 23W

Two pull tests can demonstrate a serious shortfall, but they do not define the statistical design resistance of every anchor in the school.

1.3 A condition, not a date

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · CHOICE

SCENE 42W

Halvorsen wants a date. Delacroix wants a condition. Tanaka, the hazard seismologist, points out that a date makes a promise about the ground and a condition makes a promise about the office. The next resident update is due before the office closes.

GUIDE 66W

Halvorsen and Delacroix are not disagreeing about the engineering. They are disagreeing about what the office is allowed to promise. So ask of each option one thing: is this a promise about the ground, or a promise about the office? The office controls one of those and not the other. Tanaka has already said which is which, and the update goes out before the office closes.

ASK 6W

What should the office commit to?

TAKES AS READ

a commitment can be tied to a date or to an event · peak ground acceleration and what a building feels — taken as read · p and S waves, and what the gap between them measures — taken as read

VERDICT 87W

A date promises something about the ground that the office cannot control. A condition promises what evidence the office will require before changing the restriction. Publishing those conditions turns an open-ended wait into visible progress. Residents can see what remains: utility clearance, access checks, targeted assessment or repair. The aftershock rate is a poor release trigger because it does not measure the local hazard on a specific street. The commitment should therefore name the observations and actions that will justify reopening, not a day selected for reassurance.

TAKEAWAY 18W

Tie a decision to a condition somebody can check, and it stays honest whatever the ground does next.

END OF THE DAY 15W

The earthquake is one event; the shaking a building receives depends on where it stands.

DAY 2 OF 15

The number everybody has already heard

PLAN CARD 128W

Saturday, and the town has two numbers for one earthquake. One agency said 6.6 within 20 minutes. Another said 6.8 six hours later. Rei Tanaka, the seismologist, can explain the gap to an engineer in one sentence. She has spent the morning failing to explain it to a radio station. Funmi Adeyemi, the public information officer, has to put something in a notice by four. Meanwhile Inês Cardoso, on the seismic network, has found a second problem. Two field stations have been running on a clock that drifted since spring. That moves where the epicentre plots on the map. So the office has to fix the location first. Only then can it defend the size. Today you write down what the difference between 6.6 and 6.8 actually means.

BRIEFING 14W

Two agencies published two magnitudes, and the difference is being read as a disagreement.

WHAT YOU DECIDE 19W

Separate source size and location from local shaking before the town turns two preliminary numbers into two different earthquakes.

2.1 A number, with a method attached

MATERIALS & TESTING · A ROOM · BALLPARK

SCENE 37W

4 cores, 100 mm across, cut from the cracked columns. Sørensen caps and crushes them: 28, 31, 24 and 30 megapascals. The 1974 drawing calls for 25. The four labelled specimens are still on the testing bench.

GUIDE 58W

Work the third core's strength out from the machine's own numbers rather than reading it off the report. Stress is the crushing force divided by the area the force acted on, and the area is the circle the core presents — not its diameter. One tile is that diameter, left for somebody who divides by the wrong thing.

ASK 18W

The third core failed at 244 kN. What compressive stress is that, and what does it actually settle?

BEHIND THE BUTTON 111W

Why area matters. Compressive stress is load divided by loaded area. A 100 mm diameter core presents a circular area of about 0.00785 m^2; dividing by the diameter instead of the area gives the wrong units and the wrong answer.

What the peak load tells you. The maximum compressive load divided by area gives the specimen's measured compressive strength for this simplified exercise. Concrete does not have a sharp metal-like yield point that this stop needs to identify.

What a core cannot settle. A column's residual capacity also depends on reinforcement, confinement, axial load, damage geometry and the load path. Those do not travel with a concrete cylinder into the press.

TAKES AS READ

concrete strength is measured by crushing a sample of known size

VERDICT 83W

Stress is $\sigma=F/A$. The 100 mm diameter core has area $\pi(0.05\ m)^2\approx0.00785\ m^2$, so 244,000 N / 0.00785 m^2 ≈ 31 MPa. That is a specimen compressive-strength result. The recorded shortening can be part of a stress–strain curve, but the ratio at failure is not being treated here as a yield modulus. More importantly, the core says nothing direct about reinforcement, confinement, axial load, crack mechanism or residual member capacity. Four reasonable core results narrow the diagnosis; they do not clear the columns.

TAKEAWAY 16W

Core crushing tests characterize sampled concrete strength; the damaged column still requires a member-level structural assessment.

2.2 Why the number moved

HAZARD & FORECASTING · INCIDENT BASE · A PERSON · BALLPARK

SCENE 31W

Tanaka has both agency bulletins on the desk. The early one used the first stations to report; the later one used the whole network and the long-period part of the record.

GUIDE 69W

The later solution has a fault model behind it: an area and an average slip. Build the seismic moment from the rigidity, the area and the slip, and the verdict turns that into a magnitude. Watch the units — the area is given in square kilometres and the moment needs square metres, so one of the tiles is the conversion already done and another is the same area unconverted.

ASK 15W

From the later solution's fault model, what seismic moment does the 6.8 estimate correspond to?

BEHIND THE BUTTON 162W

What the moment is made of. Rigidity times ruptured area times average slip. It is a physical quantity with units of newton metres, and it is what a long-period record measures — which is why a bulletin written from the first few seconds of short-period data can be revised once the whole waveform has arrived.

Why the scale is logarithmic. Magnitude is two thirds of the base-ten logarithm of the moment, less a constant, so one step of 0.2 in magnitude is a factor of about two in moment. Nothing about the earthquake changed between the bulletins; the estimate of how much fault moved by how much did.

Why this matters for the notice. A town told the number went from 6.6 to 6.8 hears that the earthquake grew. What actually happened is that the second estimate saw twice as much moment as the first one could, and the difference is a property of networks and waveforms rather than of the ground.

TAKES AS READ

a magnitude is computed from records, so more records can change it · magnitude as a logarithmic scale — taken as read

VERDICT 134W

Seismic moment is $M_0 = \mu AD$ — rigidity times ruptured area times average slip. The later solution's model is 600 km^2 of fault, which is 6.0×10^8 m^2, slipping an average of 1.1 m through crust with a rigidity of 3.0×10^{10} Pa, so $M_0 = 3.0 \times 10^{10} \times 6.0 \times 10^8 \times 1.1 \approx 2.0 \times 10^{19}$ N·m. Moment magnitude is $M_w = ⅔ \log_{10} M_0 - 6.06$, and $⅔ \times 19.30 - 6.06 = 6.80$. The first bulletin's 6.6 is $10^{(1.5 \times 0.2)} \approx 2$ times smaller in moment, which is exactly what a rapid short-period estimate misses: the long-period part of the record is where a large slow rupture shows up. So the revision is a statement about network coverage and waveform bandwidth, and the fault moved once.

TAKEAWAY 21W

> A magnitude revision means the estimate changed as the available waveform and network coverage improved; the earthquake itself did not change.

2.3 A notice somebody acts on

PUBLIC SAFETY · A ROOM · SEQUENCE

SCENE 30W

Adeyemi has 200 words and an audience that has heard 2 numbers and a rumour about a bigger one coming. She wants to know what the notice has to contain.

GUIDE 70W

All four of these will be in the notice, so nothing is being left out. Ask instead which one somebody can act on while standing in the street, and which is a fact they will repeat later. Two hundred words are read in the order they are printed. A revision from 6.6 to 6.8 changes nothing about what to do outside this building, and a rate is not a prediction.

ASK 9W

Say it in a way that survives being repeated

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a public notice is read once, quickly, by somebody who is tired · p and S waves, and what the gap between them measures — taken as read

VERDICT 78W

People act on the first thing they can use, so the action goes first. The reason comes next because it helps the instruction survive retelling. The magnitude revision belongs after the immediate action and explanation. It matters scientifically, but it does not change what someone should do outside this building. The aftershock forecast comes last and must be framed as a rate. That keeps a useful probability statement from being repeated as a prediction of a particular event.

TAKEAWAY 18W

A notice earns its place by changing what somebody does, and everything else in it competes with that.

END OF THE DAY 15W

Magnitude describes the source; local damage depends on distance, ground and the structure as well.

What the placard is claiming

PLAN CARD 129W

Sunday, and the first placards are being argued about. Alan Whitcombe, who leads the second assessment team, got through 400 buildings in 2 days. That is fast. He is the first to say what the speed bought and what it cost. A green placard on a house in the Flats means one engineer stood outside it for six minutes. Yvette Delacroix, the ward councillor for the Flats, has residents asking her whether green means safe. Peter Ives, who manages the hospital's facilities, wants to know why his yellow card has not moved in three days. Neither of them is being unfair. The question under both is what a rapid assessment can actually see. Today you decide which buildings have to be looked at again, properly and from the inside.

BRIEFING 18W

400 buildings were placarded in 2 days by 6 people. Somebody has to say what that is worth.

WHAT YOU DECIDE 19W

Decide what a rapid placard actually claims before residents treat green as a promise that nothing can be wrong.

3.1 A measurable acceptance target

MATERIALS & TESTING · A ROOM · BALLPARK

SCENE 38W

Ferreira, the materials engineer, has cone penetration results from a trial panel: the fill tested at 4 MPa before the stone columns went in and 11 after. The two CPT traces lie side by side on the desk.

GUIDE 58W

Five numbers, and three belong elsewhere: the depth of the fill, the amplification factor and a concrete strength. Ask of each whether it came from these two cone traces. And note what the factor is and is not. Cone resistance is an input to a liquefaction assessment, so 2.75 times the resistance is not 2.75 times the margin.

ASK 9W

By what factor did the measured cone resistance increase?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

CPT stands for cone penetration test: a standard cone is pushed into the ground and its resistance is recorded

VERDICT 84W

The measured cone resistance rose from 4 to 11 MPa, a factor of 2.75. That is a strong, auditable construction-control result: the treated ground is substantially more resistant to penetration at that location. But qc is an input to liquefaction evaluation, not a direct safety factor. Cyclic resistance also depends on overburden, fines, groundwater and the design method used. A good specification can therefore require a target corrected CPT resistance or verified performance profile without claiming that '2.75 times qc' means '2.75 times safer.'

TAKEAWAY 26W

CPT resistance is a measurable quality-control result; the acceptance criterion still has to be tied to a liquefaction evaluation rather than a vague “times safer” claim.

3.2 Green is not a guarantee

PUBLIC SAFETY · A PERSON · CHOICE

SCENE 32W

Delacroix has three residents outside with green placards and one question between them: is my house all right? Adeyemi will not say yes and needs a sentence that is not a dodge.

GUIDE 60W

All four options are things a resident might hear as reassurance. They differ in how much they claim. Ask of each whether the rapid inspection could have established it. Three residents are outside asking whether their house is all right. A sentence that promises more than the inspection did is not kindness, and the placard will be quoted back later.

ASK 11W

What does a green placard allow you to tell a resident?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

people read a colour as a verdict

VERDICT 73W

A rapid post-earthquake placard communicates the result of an evaluation, not a guarantee about every concealed part of a building or every future aftershock. The office can say that the inspected scope did not show a condition requiring restriction. It must also preserve the limits of that scope. That distinction becomes operationally important at the school, the hospital and the Ferry Street basement, where the first team recorded spaces it could not inspect.

TAKEAWAY 20W

> A placard is an occupancy decision based on a defined inspection, not a warranty that every hidden part is undamaged.

3.3 The ones worth a second visit

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · TRIAGE

SCENE 36W

The field roster locks in ten minutes. Two engineers are free tomorrow, four addresses are on Tanaka's board, and each has somebody waiting for an answer. Once the roster closes, the other three wait another day.

GUIDE 62W

All four buildings deserve an answer and only two engineers are free. Ask of each two things: what did the first visit fail to see, and what is the cost of being wrong. An empty red building is already restricted, so a second visit changes little today. An anxious owner needs a conversation, which is not the same resource as an engineer-day.

ASK 7W

Which building gets the detailed evaluation first?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

a second inspection is a scarce thing to spend

VERDICT 83W

A second inspection is valuable when the first one left an important condition unseen and the consequence of being wrong is high. The hospital is worst on both counts. 90 patients are inside, and the unresolved plant room is exactly the item preventing a clearer placard. The red warehouse is empty and already restricted, so another visit changes little today. The two green buildings have lower consequence. Their owners still deserve answers, but that can begin with communication rather than a scarce engineer-day.

TAKEAWAY 18W

> Re-inspect where the first look was least able to see and the consequence of being wrong is largest.

END OF THE DAY 15W

A placard is only as broad as the inspection behind it; unseen spaces stay unresolved.

The building that looks worst

BEFORE THE DAY — TRIAL

The Flats open today, and the vehicles come out with them

Everything across the scarp has been walkable all week and nothing out there has been called. That changes today: the fill, the wharf and Ferry Street go on the list, and they are far enough out that the van is signed out to reach them. Drive the route once before the calls start sending you down it.

PLAN CARD 113W

Monday, and Marina Court is on the front page. Six storeys of flats now stand about eight degrees out of plumb above the marina. The frame has not racked; the raft and superstructure rotated together. A survey across the roughly twelve-metre foundation finds about 1.7 metres of elevation difference from one edge to the other, with fresh sand ejecta concentrated toward the low side. Bram Halvorsen wants the block demolished this week because the lean looks like the town's obvious structural failure. Elena Navarro wants the cordon kept but the next five investigation-days spent on the reclaimed ground under the twelve neighbouring buildings. Before either decision, you have to identify what actually failed.

BRIEFING 18W

Marina Court looks like the obvious structural failure. The first question is whether the structure failed at all.

WHAT YOU DECIDE 19W

Work out why Marina Court tilted as one piece and decide what evidence matters for the twelve neighbouring buildings.

4.1 The raft is fine and the building is not

GEOTECHNICAL · A ROOM · CHOICE

SCENE 46W

Navarro's survey: the raft is intact and the frame is still square to itself, but the whole building has rotated. Across the roughly 12 m raft, one edge now sits about 1.7 m lower than the other. Fresh sand ejecta are concentrated toward the low side.

GUIDE 37W

The geometry has to agree with the explanation. An eight-degree rigid-body tilt across about twelve metres requires roughly $12 \times \tan(8°) \approx 1.7$ m of elevation difference. The frame has not racked; the support beneath it moved.

ASK 5W

Which explanation fits every observation?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a foundation spreads a building's weight into the ground

VERDICT 84W

The observations point below the raft rather than to yielding inside the frame. A rigid-body rotation of about eight degrees across a twelve-metre foundation corresponds to roughly 1.7 metres of differential elevation, which matches the survey. Fresh ejecta and the concentration of movement on one side are consistent with liquefaction-related loss of bearing resistance and ground deformation. The evidence does not reveal every detail beneath an inaccessible raft, so this remains a strong geotechnical inference rather than a direct view of the failure surface.

TAKEAWAY 10W

A structure can fail without anything in the structure breaking.

4.2 What eight degrees does

STRUCTURAL ASSESSMENT · A PERSON · CHOICE

SCENE 38W

Okonkwo has the column loads. At eight degrees the weight above each column no longer lands over its centre, and the offset grows with height up the building. The survey sheet and column plan are open beside him.

GUIDE 45W

The frightening part is the angle, but the engineering question is what that angle does to the load path. At eight degrees, gravity loads act with large eccentricity, increasing bending and second-order P–Δ demand. Continued settlement is a separate geotechnical hazard and must be monitored.

ASK 10W

What is the engineering objection to leaving Marina Court standing?

BEHIND THE BUTTON 195W

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TAKES AS READ

a leaning column carries its load off-centre

VERDICT 97W

Once the whole building is tilted, gravity no longer acts through the same geometry for which the columns and frame were proportioned. The lateral offset creates additional bending and second-order P–Δ effects, and an eight-degree lean is far beyond something that can be dismissed because the members are visibly uncracked. The survey therefore supports a red restriction while engineers check global stability, member demand and whether the ground has stopped moving. The angle does not guarantee that the lean will keep increasing on its own; renewed settlement or aftershocks are separate mechanisms that have to be monitored.

TAKEAWAY 19W

> The lean matters because gravity now acts with eccentricity, creating P–Δ effects and extra bending demand in the frame.

4.3 The same event, recorded twice

SEISMIC NETWORK · A ROOM · CHOICE

SCENE 47W

Inês Cardoso, the network technician, has the vault record from the bench and the field station record from Bay Road. Same event, same minute, 1100 metres apart. One trace is short and sharp, the other is three times taller and goes on for half a minute longer.

GUIDE 64W

All four options explain two records that disagree. They differ in what they blame: the instrument, the ground, the distance, or the timing. Two of them are checkable from the scene. Same event, same minute, 1100 metres apart. Ask what 1100 metres can and cannot account for. Which one is right decides whether the Flats get a stronger design value or a service call.

ASK 9W

What does the difference between the two records mean?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

an instrument records the shaking where it stands, not the earthquake itself

VERDICT 100W

Both instruments recorded the same earthquake, so the source is not the variable. In the band shown on the board, the Flats amplitude is about three times the reference-station amplitude. That is direct evidence that the shallow ground is changing the motion before it reaches the surface. Soft, low-velocity deposits can amplify some frequencies strongly, while other frequencies may be amplified less or even deamplified. The useful conclusion today is therefore not 'the Flats always shake

> three times harder'; it is 'the ground beneath the Flats matters enough that site response has to be measured and carried into the reassessment.'
>
> **TAKEAWAY** 21W
>
> A soft-soil/rock ratio from one band and one event is evidence of site response, not a universal multiplier for every structure.

4.4 A notice somebody acts on — Review

PUBLIC SAFETY · A ROOM · CALLBACK · SEQUENCE

SCENE 40W

A magnitude 5.1 aftershock woke the town at ten past four. Funmi Adeyemi, the public information officer, has a radio slot at seven and four things she could put in it. The phones have been going since half past four.

GUIDE 73W

All four of these will be in the slot, so nothing is being cut. Order them instead. Ask of each whether somebody hears it at seven o'clock and does something differently, or whether it is a fact they will repeat to a neighbour later. Then ask which one stops a rumour rather than feeding it. A radio slot is heard once, in the order it is read, by somebody who did not sleep.

ASK 10W

Order what the seven o'clock slot says about an aftershock

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a magnitude scale counts in factors rather than in equal steps · a public notice is heard once, quickly, by somebody who is tired

VERDICT 92W

The order is what somebody can use. The instruction goes first, because a person listening at seven acts on the first usable sentence and may not hear the rest. The reason follows, since a reason is what lets an instruction survive being retold. The comparison comes third: each unit of magnitude is about 32 times the energy, so 1.7 units of it is a factor near 350, and that number is what stops a 5.1 being heard as the start of something worse. The forecast is last, and it is a rate.

TAKEAWAY 26W

A magnitude scale counts in factors, so the gap between two numbers is a ratio, and the ratio is the part a frightened listener actually needs.

END OF THE DAY 17W

A building can be badly unsafe because its support moved even when its frame did not fracture.

DAY 5 OF 15

The one everybody assumed was fine

PLAN CARD 118W

Tuesday, and the school is the quiet problem. Bay Road School came through with a green placard. Its classrooms look untouched. The district would like 400 children back in it next week. Alan Whitcombe placarded the school on day one from the street outside. On his second list he wrote that the gymnasium was locked. That gym is a single-storey hall with a long-span roof. It has no internal walls. Its side walls are heavy precast panels. Nobody has been inside it since the quake. Marisol Okonkwo has the keys this morning. You need to know what holds walls like those to a roof like that. Today you decide whether 400 children go into that hall on Monday.

BRIEFING 17W

Bay Road School is green, term starts in a week, and nobody has been inside the gym.

WHAT YOU DECIDE 19W

Re-open a green school before 400 children return, and find the load-path failure a street inspection could not see.

5.1 A hall with nothing in the middle

STRUCTURAL ASSESSMENT · A ROOM · CHAIN

SCENE 36W

Bay Road School's gym has a long roof, heavy precast wall panels and no interior bracing line. The structural drawings are pinned across a table. One panel connection tested poorly, but the larger members look undamaged.

GUIDE 61W

Build the path the sideways force actually takes, in order, from the moving mass to the ground. Then name the one required transfer that would break the path if it went. The largest member is not automatically the answer: a path is only as good as its weakest handover, and the readings on each link tell you which handover that is.

ASK 12W

Which link governs the panel load path when the building shakes sideways?

BEHIND THE BUTTON 176W

What a lateral system is. In an earthquake the mass of the building wants to keep going sideways while the ground moves under it. That inertia has to be carried by something, handed on, and eventually delivered to the foundation. Each of those handovers is a connection, and a connection that cannot carry its share ends the path there.

Why a gym is the hard case. A long roof with heavy precast panels and no interior bracing line has very few paths available: the roof has to act as a horizontal plate and hand its load to the end walls, and the panels have to be tied into that plate. Remove one tie and the panel is a free-standing wall carrying its own inertia.

Why undamaged members are not reassurance. The beams and columns can be in perfect condition while the connection between them is not, and a connection is exactly what the drawings say least about. One panel connection tested poorly, which is a statement about the class of connection rather than about that panel.

TAKES AS READ

a horizontal force has to travel through a building to reach the ground

VERDICT 85W

Earthquake inertia does not disappear because the visible structural members are large. A base-shear relation such as $V = C_sW$ estimates a total lateral demand, but that force still needs a continuous path to the ground. For the gym panel, inertia crosses the panel-to-roof tie, enters the roof diaphragm, reaches the resisting end wall or frame, and then reaches the foundation. The small tie is therefore load-bearing in the literal sense: if it opens, the rest of the strong path cannot collect that panel's force.

TAKEAWAY 19W

A lateral system works only when every required transfer carries force continuously from the moving mass to the foundation.

5.2 Ground that people put there

GEOTECHNICAL · A ROOM · PROTOCOL

SCENE 34W

Navarro has an 1892 survey and a photograph from 1948. Where Bay Road now runs there was a tidal creek, and the port was pumped into place out of the dredger over four summers.

GUIDE 57W

Four descriptions of ground, and four behaviours under shaking. Pair them by asking two things of each: how loose is it, and can the water get out. Age is not the variable it looks like. One of these is the wettest and loosest of all. It is a line on an 1892 survey, invisible from the street.

ASK 10W

Say what the ground under the Flats is made of

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

ground can be natural or placed by people

VERDICT 79W

Hydraulic fill is deposited loose and commonly remains saturated. During shaking, grains try to rearrange while water cannot escape quickly. Effective stress is $\sigma' = \sigma - u$: as pore-water pressure u rises, the grain-to-grain stress σ' falls. If that contact stress becomes very small, the soil can lose much of its shear strength and behave like a fluid. The old creek line matters because it marks especially wet, loose fill, which is also where the sand ejecta appeared.

TAKEAWAY 18W

Liquefaction needs susceptible soil, enough saturation and strong cyclic loading; young hydraulic fill often supplies the first two.

5.3 400 children on Monday

PUBLIC SAFETY · A PERSON · CHOICE

SCENE 38W

400 children are due back Monday. The classrooms are sound and the gym is not. Halvorsen has parents calling, the district has no spare campus, and the morning bus schedule depends on whatever answer comes off this table.

GUIDE 64W

All four options are defensible readings of one site with two different buildings on it. Ask of each whether it treats the site as one thing or as parts. The classrooms and the gym have different evidence behind them. Four hundred children arrive Monday and there is no spare campus. A decision that covers both buildings equally is ignoring half of what was measured.

ASK 9W

What should happen to Bay Road School on Monday?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a building can be used in part

VERDICT 84W

The useful decision is component-specific. The classroom block has no identified structural deficiency, while the gym has heavy panels attached by connections whose tested capacity is below the estimated demand. If the gym can be physically isolated from classroom access and egress, the classrooms can reopen while the hall remains closed and the connections are repaired. If routes or falling hazards overlap, that separation has to be solved first. Partial occupancy is not optimism; it is a controlled use tied to the actual defect.

TAKEAWAY 17W

> The useful question is which parts can be occupied, not whether the site is open or shut.

5.4 The same event, recorded twice — Review

SEISMIC NETWORK · A ROOM · CALLBACK · PROTOCOL

SCENE 37W

Inês Cardoso, the network technician, has four records of one aftershock from four instruments around the bay. Same event, same minute. The traces disagree about how tall they are and about how long the shaking went on.

GUIDE 89W

Four instruments, one event, four records that disagree. Two of the sites are the same short distance from the source and two are ten times further out. Pair them by asking two things of each site in turn: how far has the wave travelled, and what is it travelling through at the end of the journey. Distance takes amplitude away across the whole band at once. Ground gives some of it back in a band of its own. One of these sites is far away and still not quiet.

ASK 8W

Pair four instruments with what each one recorded

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an instrument records the shaking where it stands · distance costs a wave amplitude on the way

VERDICT 92W

Magnitude belongs to the source and there is one of it. What an instrument writes down is the shaking where it stands, which is the source motion after spreading, after attenuation, and after whatever the last few tens of metres of ground did to it. Distance removes amplitude across the whole band. Soft ground puts some of it back in a narrow band of periods and stretches the shaking out. So a far station on

> deep sediment can beat a near station on rock at long period, and both records are honest.
>
> **TAKEAWAY** 31W
>
> There is one number for the earthquake and one record for every place, and what the last few tens of metres of ground are made of decides the difference between them.

END OF THE DAY 13W

A green exterior does not clear an interior condition that was never inspected.

DAY 6 OF 15

The hospital, and what waiting costs

PLAN CARD 106W

Wednesday, and the hospital has been yellow for four days. Peter Ives has 90 patients on the ground floor. The 2 upper floors are shut. The district's surgery is being done 60 kilometres away. Two people have already been moved who would not have been. Marisol Okonkwo wants the plant room opened and the shear walls cored before she signs. Bram Halvorsen says every day of that is a day of care not given. This morning he is right. The tests Okonkwo wants will take four days. They will not change what the building is. Today you decide what evidence, if any, is worth the wait.

BRIEFING 17W

90 patients, 3 days of yellow, and an argument about whether more evidence is worth more time.

WHAT YOU DECIDE 20W

Resolve the hospital restriction using only evidence that can still change the decision, while counting the harm caused by delay.

6.1 The part nobody has seen

STRUCTURAL ASSESSMENT · A ROOM · ATTEST

SCENE 40W

90 patients remain on the ground floor while 2 upper floors sit empty. The hospital is still yellow after three days. Frame, cladding and stair are documented. The roof plant room remains locked, with two full water tanks above it.

GUIDE 65W

Each claim on the list is signed or checked by somebody. Open the backing before you spend anything: some claims rest on an inspection of the thing they describe, and some rest on a document, an assumption or a locked door. You have one physical verification. Hold the claims the evidence does not support, and spend the visit where being wrong would cost the most.

ASK 11W

Which critical claim still needs physical verification on the current building?

BEHIND THE BUTTON 162W

What a record is and is not. A signature says somebody took responsibility for a statement. It does not say the condition was observed, and on a building three days after an earthquake those two come apart constantly: drawings are checked, cladding is looked at from the street, and a locked plant room is written up from the last inspection before the shake.

Why the tanks matter. Two full water tanks are mass, high up, in a room nobody has entered since the earthquake. If their supports have moved, the consequence lands on the floor below — where ninety patients are. The claim about that room is therefore the one whose being wrong costs the most, whatever its paperwork says.

Why one check rather than four. Verification costs time the hospital does not have while it stays yellow. Choosing which claim to check is the actual professional judgement, and it is a judgement about consequence rather than about which record looks flimsiest.

TAKES AS READ

a building's parts can be cleared separately

VERDICT 82W

A yellow placard should reduce to specific unresolved conditions, not a general feeling that more inspection is always safer. The frame, cladding and main stair already have direct observations behind their claims. The roof plant room does not. Its full water tanks add substantial mass high in the building, and nobody has seen their restraints. With one verification visit, repeating a comfortable backed inspection wastes the chance to resolve the one critical claim that still lacks physical evidence on the present configuration.

TAKEAWAY 22W

A signed or checked record is useful only to the extent that it is backed by observation of the condition it claims.

6.2 What four more days buys

HAZARD & FORECASTING · INCIDENT BASE · A PERSON · CHOICE

SCENE 37W

Tanaka has the aftershock rate for the coming week. Ives has the transfer log. Halvorsen wants both of them stated as consequences rather than as principles. The review board meets again at the end of the week.

GUIDE 61W

All four options are ways of weighing four days. They differ in what they count. Ask of each whether it prices the delay, the evidence, or neither. The frame has already been inspected and the unresolved item is a roof plant room nobody has entered. So the real test is what this evidence could change, and coring answers a different question.

ASK 10W

How should the office weigh four more days of evidence-gathering?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a delay has consequences that can be counted

VERDICT 89W

Both sides of the decision have consequences and both can be estimated. Delay carries transfers, postponed care and operational strain. Additional testing is worthwhile when it has a realistic chance of changing the safety decision. Here the frame has already been inspected and the unresolved item is the unvisited roof plant room with heavy tanks; a four-day coring programme answers a different question. The disciplined test is therefore value of information: what decision could this evidence change, how likely is that, and what is paid while waiting for it?

TAKEAWAY 22W

> A decision to wait is a decision with an outcome, and it has to be compared with the outcome of deciding now.

6.3 The instrument in the basement

SEISMIC NETWORK · A ROOM · BALLPARK

SCENE 51W

There is an accelerograph in the hospital basement, installed in 1998 and finally useful. Cardoso has processed the record into a 5%-damped response spectrum. Around the hospital's roughly 0.4 s fundamental period, the spectral acceleration is about 0.31 g. The comparable screening coefficient used for the old design check is 0.35.

GUIDE 42W

Do not use the raw peak ground acceleration as though the whole building were a rigid block. Use the period-specific spectral ordinate already read from the record, then apply the same simplified V = C·W screening relation used in the design comparison.

ASK 19W

Using the period-specific spectral coefficient, what simplified base-shear demand does the record imply, compared with the old design check?

BEHIND THE BUTTON 104W

What the instrument gives you. An accelerograph records acceleration versus time. From that record, engineers can compute a response spectrum showing the demand on idealized oscillators at different periods.

Why period matters. A multi-storey building does not necessarily respond at the instant of the record's largest ground acceleration. The spectrum near the building's period is a better screening input than raw PGA for comparing dynamic demand.

What the comparison still cannot prove. A simplified spectral base-shear screen does not inspect the plant-room restraints, duration effects, higher modes, irregularity or damaged details. A value below the historical design comparison is useful evidence, not a clearance.

TAKES AS READ

a building can be instrumented, and the record is about that building · peak ground acceleration and what a building feels — taken as read · peak ground acceleration and what a building feels — taken as read

VERDICT 99W

For this screening calculation the same simplified relation is used on both sides: V = C·W. The processed record gives C ≈ 0.31 at about 0.4 s and the hospital's seismic weight is about 85 MN, so V ≈ 0.31 × 85 ≈ 26 MN. The historical design-screen coefficient 0.35 gives about 30 MN. The useful result is a like-for-like comparison of a measured period-specific demand with the old screening check. It is not proof that every component remained below capacity, and it is deliberately not obtained by treating raw PGA as the force coefficient for the entire building.

TAKEAWAY 25W

A building record is most useful when converted to the response quantity the structure actually uses; raw PGA is not automatically the building base shear.

6.4 The frame, or the finish

STRUCTURAL ASSESSMENT · A ROOM · BELT

SCENE 42W

Four hundred photographs came in from the rapid assessment and the panel sits at four. Some of what is in them is the structure. The rest is what was hung on it, and the two produce the same alarm in a caption.

GUIDE 47W

Two bins. Structural damage is to the things carrying load down to the ground — columns, beams, walls that hold something up, connections between them. Non-structural damage is to what the building carries: cladding, ceilings, partitions, services. Sort on whether the item is in the load path.

ASK 15W

Send each photograph to the bin that says whether it is in the load path.

BEHIND THE BUTTON 82W

Why the split decides the placard rather than the repair bill. A cracked partition is expensive and does not change whether the building stands up in the next aftershock. A cracked column does, and it can be behind a partition that looks fine.

Why non-structural damage still kills people. Falling cladding, a collapsed ceiling grid and a toppled bookcase all injure. They are dealt with by clearing and securing rather than by shoring, which is a different crew and a different day.

TAKES AS READ

some parts of a building hold it up and some are carried by it

VERDICT 149W

The placard question is whether the building will stand up to the next shake, and only what carries load answers it. Columns, beams, shear walls and their connections are the path down to the foundation, and damage to any of them changes capacity. A shattered partition, a collapsed ceiling grid, cladding hanging off a facade and a toppled bookcase are all carried *by* that path, and none of them makes the frame weaker. That is why the sort matters more than it sounds: the alarming photographs are mostly non-structural, and the photograph that decides the placard is often a hairline diagonal on a column behind an intact wall. Non-structural damage still injures people — falling cladding is the commonest injury in a moderate earthquake — so the answer is not that it does not matter. It is that it wants a different crew, a different urgency, and no shoring.

TAKEAWAY 14W

What decides occupancy is what is in the load path, not what looks worst.

END OF THE DAY 21W

More evidence is valuable only when it can change a decision enough to justify the time and cost of getting it.

DAY 7 OF 15

What the ground will do next

PLAN CARD 100W

Thursday, and the question in every room is when the aftershocks stop. Rei Tanaka has the sequence. There have been 210 events above magnitude 3 in 4 days, and the count is falling steadily. Bram Halvorsen wants a date he can put on a notice. Funmi Adeyemi has to write that notice. Marisol Okonkwo needs the same rate for a different reason. A damaged building can carry less than it could. A red placard is urgent rather than tidy because that weakened building may be shaken hard again. Today you say what the forecast is, and what it is not.

BRIEFING 12W

41 aftershocks yesterday, 26 today. Everybody wants to know when it stops.

WHAT YOU DECIDE 23W

Turn the aftershock sequence into a forecast the inspection teams can actually use without pretending it predicts a date or the next event.

7.1 The rate, falling

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · BALLPARK

SCENE 30W

Day 1 had 96 events above magnitude 3. Day 2 had 52, day 3 41, day 4 26. Tanaka's fit has the rate roughly halving as the elapsed time doubles.

GUIDE 63W

Five numbers, and two of them belong to other questions: the magnitude and the running total. One is day one, which is four doublings back rather than one. Ask of each whether it is a count, a time, or a ratio of times. And note what comes out: an expected rate, not a quota. Real days scatter either side of a fitted curve.

ASK 10W

Estimate the number of magnitude 3+ aftershocks on day 8.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate can fall smoothly without ever reaching zero

VERDICT 68W

The exercise uses a simple Omori-style $p \approx 1$ fit. If the rate is about 26 M3+ events per day around day 4, doubling the elapsed time to day 8 gives a first-order expectation near 13 per day. Real operational forecasts carry uncertainty and are updated as the sequence evolves. Thirteen is therefore a central rate estimate for planning, not a promise that exactly thirteen events will occur.

TAKEAWAY 23W

Aftershock rates often decay roughly as a power law; the fitted rate is an expectation, not the number the next day must contain.

7.2 A rate is not a schedule

SEISMIC NETWORK · A PERSON · BALLPARK

SCENE 36W

The notice draft says the aftershocks will have stopped by the end of the month. Cardoso points at the sentence and asks where the month came from. The sentence is due for the afternoon public bulletin.

GUIDE 64W

Put the fitted numbers through the decay law and see what day thirty actually gives. The rate falls as a power of the time since the mainshock, so pick the productivity, the time offset and the elapsed days, and read the answer as events per day. One tile is the exponent, which goes in the power and not in the denominator on its own.

ASK 17W

At day 30 after the mainshock, what does the fitted decay law give as the expected rate?

BEHIND THE BUTTON 180W

What the law says. The rate of aftershocks per day falls roughly as one over the elapsed time raised to a power near one, with a small offset that keeps the first hours finite. It is a decay, not a countdown: the curve approaches zero and never arrives, which is why no end date can be read off it.

Why a fitted rate is not a schedule. The number that comes out is an expectation over an interval — so many per day on average. Actual counts scatter around it, some days three and some days none, and a rate of a tenth of an event per day does not promise the tenth day is the one.

Why size is a separate question. The decay law describes how often, and the magnitude distribution describes how large. A falling rate lowers the chance of a large aftershock in the same proportion as a small one, and rules nothing out. Publishing a month is therefore two mistakes: an end where there is a fade, and a date where there is a rate.

TAKES AS READ

a forecast states what is expected on average · aftershock decay as a power law — taken as read

VERDICT 62W

With the fitted toy model $n(t)=84/(0.2+t)^{1.05}$, the day-30 instantaneous rate is about $84/30.2^{1.05} \approx 2.4$ events per day above M2. The model never supplies an 'end date'. A useful forecast asks about a future interval—tomorrow, the next week, the next month—and turns the fitted rate plus model uncertainty into expected counts and probabilities. That is the quantity an inspection schedule can use.

TAKEAWAY 18W

An Omori–Utsu rate approaches zero gradually; operational forecasts integrate that rate over a future window and report uncertainty.

7.3 Damaged, and waiting to be asked again

STRUCTURAL ASSESSMENT · A ROOM · CHOICE

SCENE 37W

Okonkwo has 11 buildings with cracked shear walls that are still standing. Each has less capacity than it had a week ago, and the sequence has not finished. The shoring list is pinned beside the new forecast.

GUIDE 60W

All four options give a reason the eleven buildings stay urgent. They differ in what is changing: the hazard, the buildings, or both. Ask of each whether the falling aftershock rate contradicts it. Two of these say the shaking will get worse, and the forecast says otherwise. What has changed since Friday is on the other side of the comparison.

ASK 12W

Why does a falling aftershock rate still leave these 11 buildings urgent?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a damaged element carries less than it used to

VERDICT 83W

The hazard is declining, but the buildings are not the same buildings they were before the mainshock. Cracked shear walls may have lost capacity, so a smaller aftershock can now produce additional distress. That does not mean every magnitude-5 event will damage them or that the exact remaining capacity is known. It means the acceptable demand has changed. The 11 buildings therefore stay urgent because the sequence still supplies repeated loading while their structural reserve is reduced and has not yet been restored.

TAKEAWAY 18W

> Damage lowers the capacity, so the same aftershock that was survivable last week may not be this week.

7.4 Ground that people put there — Review

GEOTECHNICAL · A ROOM · CALLBACK · CHOICE

SCENE 40W

Thandi Mbeki, the geotechnical field technician, has photographed two streets 300 metres apart on the same fill. Ellis Street is grey with sand fans, kerb to kerb. Tarrant Street, on the same reclaimed ground and the same shaking, has none.

GUIDE 91W

Both streets sit on the same hydraulic fill at the same distance from the source, and the two nearest instruments agree on what the shaking was, so the shaking is not the variable. Navarro's boreholes find the same loose sand at two to five metres under each of them. The one number the two logs disagree on is the depth of the water table: 1.5 metres under Ellis Street and 7 metres under Tarrant Street. Three conditions have to arrive together. Ask which option names the one Tarrant Street is missing.

ASK 10W

Why did one street liquefy and the neighbouring one not?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

ground can be loose without being wet

VERDICT 88W

Liquefaction wants three things together: a soil loose enough to want to pack down, enough water in the gaps that it cannot pack down without raising the pore pressure, and shaking strong enough and long enough to drive it. It is the rising pore-water pressure that takes the grain contacts apart and leaves the layer with almost no shear strength to offer. Under Tarrant Street the loose layer sits above the

> water table, so there is no pore pressure there to raise. Same fill, same shaking, dry street.
>
> **TAKEAWAY** 24W
>
> Susceptible soil, saturation and strong enough shaking are three separate conditions, and a street that fails any one of them stays where it is.

END OF THE DAY 13W

An aftershock forecast is a probability distribution over future events, not a schedule.

The material, and what it can still carry

BEFORE THE DAY — FOLLOW

Mbeki walks the liquefaction line once

The boundary between the granite bench and the fill is not on any map at the accuracy anybody needs, and Mbeki, the geotechnical field technician, can see it in the ground. She walks it once, this morning, stopping where the surface changes. Stay with her — far enough back to see what she is pointing at, close enough not to lose which way she turned at the ejecta.

PLAN CARD 109W

Friday, and the argument at the base has moved from buildings to numbers. The public library is a 1970s concrete frame. Three of its ground-floor columns have diagonal cracks. Three engineers have looked at those columns. One called the cracking moderate, one serious, one superficial. Duarte Ferreira, the materials engineer, points out that none of those words is a number. His laboratory can give the office a number in a day. Kirsten Sørensen has already cut four cores for him. She refused a fifth. It arrived with no label saying which column it came from. Today you decide what the library may be used for while the testing runs.

BRIEFING 13W

The library's columns look cracked. Nobody has said what they can still hold.

WHAT YOU DECIDE 23W

Replace adjectives about the library columns with a measured material property, then keep that property separate from the capacity of a damaged member.

8.1 How far away it started

SEISMIC NETWORK · A ROOM · BALLPARK

SCENE 27W

The vault trace shows the P arrival at 04:12:19 and the S arrival at 04:12:32. Cardoso wants the distance before anybody draws a circle on a map.

GUIDE 68W

Five numbers, and two of them belong to the earthquake rather than to this station: the magnitude and the depth. Two others are clock readings, and only the gap between them carries distance. Ask of each whether it describes the source or the arrival. And note what one station can give: a radius, not a position. The circle still needs other stations before anybody draws on a map.

ASK 9W

Estimate the distance from this station to the source.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

P waves travel faster than S waves through the same rock

VERDICT 77W

Both waves leave together, but the P wave travels faster. Their arrival-time gap therefore grows with distance from the source. For shallow crust, about eight kilometres of source distance corresponds to each second of S-minus-P separation. The 13-second gap gives about 104 km. One station supplies only a radius, so the source still lies somewhere on a circle. A network combines several station distances to locate the source region rather than mistaking one distance for a position.

TAKEAWAY 20W

The gap between the two arrivals grows with distance, so one station gives a distance and three give a place.

8.2 Which way the cracks run

STRUCTURAL ASSESSMENT · A PERSON · CHOICE

SCENE 37W

The cracks run diagonally, in both directions, crossing each other in an X across the middle third of each column. They are fine, numerous and evenly spaced. Photographs of all three columns are clipped to the sheet.

GUIDE 60W

Four options, and the pattern has to be explained in full. The cracks are diagonal, in both directions, crossing in the middle third. They are fine, numerous and evenly spaced. Ask of each option what pattern it would leave instead. Crushing and settlement do not reverse direction. And note the limit: a pattern names a mechanism, not a remaining capacity.

ASK 5W

Which explanation fits the pattern?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a crack runs across the direction of the tension that opened it

VERDICT 80W

Diagonal cracking is consistent with principal tension produced by shear, and crossed diagonals fit load reversals during earthquake shaking. The pattern is therefore much more consistent with cyclic shear demand than with simple long-term settlement or axial crushing. Fine, distributed cracks can indicate that reinforcement is controlling crack widths rather than allowing one brittle split, but that is not the same as proving the column is safe. Crack pattern identifies a likely mechanism; residual capacity still needs an engineering check.

TAKEAWAY 19W

> The shape and repetition of damage can narrow the failure mechanism, but pattern recognition alone cannot establish remaining strength.

8.3 Part of a building, for part of a purpose

PUBLIC SAFETY · A ROOM · SEQUENCE

SCENE 38W

Monday's public meeting has 200 residents registered and no backup venue. The library is the only hall large enough. Three ground-floor columns are cracked, and the route plan has to be settled before the doors can be advertised.

GUIDE 66W

All four of these will happen, so the order is what is being decided. Ask of each what has to be settled before it can mean anything. A number of people is only enforceable once the permitted area is defined. A review date on a restriction is what stops it drifting into permanence. Two hundred residents are registered for Monday and there is no other hall.

ASK 9W

Decide what the building may be used for meanwhile

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

occupancy can be limited by number, by area or by time

VERDICT 83W

Partial occupancy is an engineering boundary, not a hopeful label. The meeting can use the hall only if occupied routes and seating stay outside the hazard zone. Damaged columns must also be isolated or supported as required. The temporary use must be defined in writing. A restriction saying only 'part of the building' is too vague to enforce. The plan should name the permitted area, excluded area, route and duration. That preserves usable space without silently widening the claim made by the assessment.

TAKEAWAY 21W

Restricted use is a real option between open and shut, and it has to name what the restriction is protecting against.

END OF THE DAY 14W

A concrete core measures sampled material strength; it does not certify a cracked column.

DAY 9 OF 15

The cordon nobody can lift

PLAN CARD 112W

Saturday, and the cordon around the Flats is a week old. It was drawn on day one to keep people away from falling masonry. It was kept on day three for the gas main. It is being kept today because nobody has said it can come down. Yvette Delacroix has a list of 400 households. Sixty of them are sleeping in the sports hall. She has a question the office has not answered. What has to be true before people can go home? Bram Halvorsen is losing patience. Marisol Okonkwo will not be hurried. Every street inside the line is a different case. Today you decide where the cordon should run tonight.

BRIEFING 16W

11 streets have been closed for 6 days, and the reason is now 3 different reasons.

WHAT YOU DECIDE 18W

Rebuild the cordon from the hazards that still exist instead of letting the emergency line survive by habit.

9.1 Three reasons wearing one line

PUBLIC SAFETY · A ROOM · ATTEST

SCENE 41W

Shopkeepers are waiting at the tape with deliveries and keys in hand. Delacroix walks the cordon with the boundary log. Three stretches have written reasons. The southern leg is still closed, but nobody can point to the hazard it is controlling.

GUIDE 71W

Each stretch of cordon is a claim that a hazard is there now. Open the evidence behind each one before spending anything: some rest on an inspection, some on a hazard that has since been removed, and one has no written reason at all. You can recheck one stretch physically. Hold the claims the evidence does not support, and choose the recheck where opening the wrong street would cost the most.

ASK 16W

Which current exclusion claim most needs to be checked against the physical condition on the ground?

BEHIND THE BUTTON 165W

A cordon is a claim, not a decision. Tape is put up for a reason, and the reason is a statement about a present condition — falling glass, an unstable parapet, ground that may still move. Conditions change and reasons expire, so a boundary nobody has revisited is a claim nobody has retested.

What the cost of a stale cordon is. Shopkeepers are at the tape with keys and deliveries. Every day a stretch stays closed without a hazard behind it, the cordon as a whole loses authority, and the stretches that do control real hazards get walked around. Over-closing is not the safe error it looks like.

Why the unwritten leg is not automatically the answer. A stretch with no logged reason may still be controlling a genuine hazard that somebody saw on day one and never wrote down. The point of the check is to find out, which is why the panel makes you spend the verification rather than infer from the paperwork.

TAKES AS READ

a boundary can be drawn for more than one reason

VERDICT 80W

A cordon can contain several hazards, each with its own condition for removal. The Parade segment has an overhead-masonry basis. The gas-main segment depends on utility status. Ferry Street is an access restriction for heavy emergency vehicles. The southern leg is different. The map still excludes people, but the log contains no recorded hazard supporting that claim. With one field check before the map is reissued, the unbacked segment deserves verification before a restriction with direct support is checked again.

TAKEAWAY 22W

Every cordon segment is a claim about a current hazard, and each claim should remain traceable to evidence that can be rechecked.

9.2 Soft ground, three days on

GEOTECHNICAL · A PERSON · CHOICE

SCENE 35W

Thandi Mbeki has been on Ferry Street with a hand auger. The sand fans have dried, the water table has dropped back, and the surface bears a person easily and a truck not at all.

GUIDE 62W

Four options describing the same street, from ruined to recovered. Ask of each which observation supports it and which contradicts it. The fans have dried and the water table has dropped, so the excess pressure has gone. The surface takes a person and not a truck. A dry surface says nothing about heavy loads, and drained is not the same as unchanged.

ASK 9W

What is the defensible status of Ferry Street now?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

liquefied ground regains strength as pore pressure dissipates

VERDICT 80W

Liquefaction is a transient loss of effective stress during shaking as excess pore pressure builds. After the shaking, excess pressure dissipates and grain contacts recover, but settlement, lateral displacement, ejecta, voids and heterogeneous stiffness remain. A dry surface therefore does not prove adequate bearing capacity for trucks. Susceptible saturated layers may also reliquefy in later strong shaking. The defensible placard decision is to separate what has been observed—pedestrian bearing at the surface—from the heavy-load and future-shaking questions that remain unverified.

TAKEAWAY 21W

Liquefaction is transient, but the ground left behind can have settlement, voids and uncertain bearing capacity even after pore pressure dissipates.

9.3 Where the first day of assessment goes

STRUCTURAL ASSESSMENT · A ROOM · ALLOCATE

SCENE 38W

Six assessors are free for one day. 400 displaced households are waiting on decisions in the Flats. Marina Court and the Parade are already cordoned, while Upper Town is mostly occupied. The team board shows six assessor-days available.

GUIDE 71W

Six assessor-days is the whole pool, and each package you drag in spends some of it. Watch the answers list rather than the buildings: it shows which questions the current plan can answer, and every package you buy is a question you can answer and one you have given up. The cordons already in place cost nothing. Commit the plan when the answers you can give are the ones today needs.

ASK 21W

Can your allocation preserve a same-day re-occupancy decision for the waiting Flats households while accepting that some lower-value questions will wait?

BEHIND THE BUTTON 148W

Why the pool is the constraint. Six assessors for one day is a fixed quantity of attention, and the Flats alone could absorb all of it. The decision is not what is worth inspecting — nearly everything is — but which inspections change what happens before tonight.

What makes an answer worth buying today. Four hundred households are waiting on a re-occupancy decision. An inspection that can produce that decision by this evening is worth more than one that produces a better answer next week, because the second one has already been made for them by the delay.

Why the existing cordons are on the board at zero. They are already in force, so they cost nothing to keep and they answer nothing new. They are there to be noticed: a plan that spends assessor-days re-establishing what is already closed has bought a question it had already answered.

TAKES AS READ

an assessment programme is a queue, and the order is a decision

VERDICT 81W

Assessment capacity is a finite pool, so every useful inspection has an opportunity cost. The Flats package can resolve an occupancy decision for hundreds of displaced households. Marina Court and the Parade already have protective controls in place, so extra detail there changes less today. Upper Town follow-up is useful but not essential to the immediate re-occupancy decision. The allocation should therefore preserve the Flats question first, then spend remaining capacity only where another answer is worth the assessor-days it consumes.

TAKEAWAY 18W

A finite inspection team should be spent on the questions whose answers can still change what happens today.

9.4 How far away it started — Review

SEISMIC NETWORK · A ROOM · CALLBACK · BALLPARK

SCENE 40W

Inês Cardoso, the network technician, is timing an aftershock rather than the mainshock. On the Bay Road field station the P arrival is at 21:47:06 and the S arrival at 21:47:14. She wants a distance before she calls the vault.

GUIDE 80W

Five numbers again, and the same two questions to ask of each. Does it belong to this station's record or to the earthquake itself, and is it half of a pair with a gap in it. The magnitude and the depth are both true about this aftershock and neither one is a distance. Read the seconds off the two arrivals, take the difference, and turn it into kilometres with the constant. Then ask what the answer is a radius around.

ASK 9W

Estimate the distance from this station to the aftershock.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

P waves travel faster than S waves through the same rock

VERDICT 91W

Both waves leave the source together and the P wave travels faster, so the gap between the two arrivals grows with distance. Eight seconds at about eight kilometres of source distance for each second gives roughly 64 kilometres. Neither the magnitude nor the depth is read off this measurement: magnitude comes from the amplitude of the record and depth from how the arrivals vary across the network. One station gives a radius, so the aftershock is somewhere on a circle until two more stations cut that circle down to a place.

TAKEAWAY 24W

The gap between the arrivals is a distance and nothing else, and a distance from one station is a circle rather than a place.

END OF THE DAY 20W

A cordon is a current safety decision, not a historical trace of where someone drew a line on day one.

DAY 10 OF 15

The photograph that changed the meeting

PLAN CARD 114W

Sunday, and the office is on the back foot. A resident went into the basement car park of an apartment block on Ferry Street. He photographed a cracked column down there and gave the picture to the paper. The block carries a green placard. The photograph is real and so is the crack. There are 90 flats above that basement with people back inside. Alan Whitcombe placarded the block from the street on day one. He never saw the basement. Funmi Adeyemi has a public meeting at six. Marisol Okonkwo has two hours and no key to the car park. Today you decide what to say at six about a process that missed this.

BRIEFING 17W

A resident's photograph of a cracked column is on the front page, and the building is green.

WHAT YOU DECIDE 24W

Absorb a resident photograph into the evidence chain, inspect what the green placard missed and keep the excavation safe while the answer is checked.

10.1 A real crack, and an unknown building

STRUCTURAL ASSESSMENT · A ROOM · CHOICE

SCENE 37W

The photograph has already been passed around town. It shows a basement column with a horizontal crack, pipework and no daylight. There is no scale, no building number and no way to tell which column it is.

GUIDE 66W

Four options, and they differ in how much a photograph can carry. There is no scale, no building number and no earlier picture. Ask of each option what it would need that the image does not have: a width, an age, a location, a cause. Accepting the crack as real is not the same as knowing what it means, and it is already all over town.

ASK 5W

What has the photograph established?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a photograph records what was in front of the lens

VERDICT 81W

Once the image is accepted as genuine, the crack is evidence, but its scope is narrow. The photograph has no scale, exact location, earlier comparison or useful view of the surrounding structure. It cannot establish the crack's width, age, importance or cause. It does establish that a basement area outside the rapid inspection contains a real feature worth checking. The response is therefore to locate it and gather context quickly. The photograph alone neither justifies evacuation nor supports dismissing the observation.

TAKEAWAY 15W

> An authenticated photograph can establish an observation while leaving scale, location, age and mechanism unresolved.

10.2 Down there with a torch

MATERIALS & TESTING · A PERSON · CHOICE

SCENE 37W

In the basement: the crack is 0.4 mm and horizontal, a metre above the floor. Five more columns along the row carry one that looks the same. The original construction drawings are open on a crate nearby.

GUIDE 61W

Four explanations, and one fact constrains all of them. Six columns in a row carry the same feature at the same height and the same width. Ask of each option whether it would produce that repetition. Six independent failures rarely agree to a millimetre. The drawings are open on a crate, which is where a shared construction detail would show up.

ASK 8W

What is the best-supported explanation to test first?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

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TAKES AS READ

a defect that repeats in a pattern usually has a construction cause

VERDICT 89W

Six nearly horizontal features at the same elevation make a shared construction detail a strong first hypothesis. A lift joint at the top of a concrete pour would naturally repeat from column to column. Earthquake damage can also localize at a pre-existing joint, so identifying the joint is not the same as clearing it. Check drawings or pour records, measure displacement and spalling, inspect reinforcement continuity where possible, and compare with an undamaged row. The pattern ranks the hypothesis; the follow-up inspection decides whether the earthquake changed its condition.

TAKEAWAY 26W

A repeated crack at the same construction elevation is evidence for a shared detail, but the detail must still be checked for earthquake-induced opening or slip.

10.3 The meeting

PUBLIC SAFETY · A ROOM · SEQUENCE

SCENE 40W

200 people are already in the hall. One photograph has been forwarded all over town. The office has to explain what it found tonight, what it did not see on day one, and what happens next before anyone goes home.

GUIDE 66W

All four of these have to be said tonight, so nothing is being withheld. Ask which one the hall came to hear, and which one it will only accept afterwards. Two hundred people are already seated and the photograph has been forwarded all over town. An explanation offered before the finding reads as a defence, and the admission is the part that has to be plain.

ASK 5W

Say it at six o'clock

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

trust is affected by how a process handles being wrong

VERDICT 88W

The finding goes first because that is what everyone came to hear. Its meaning follows: the repeated horizontal crack fits a construction joint better than new earthquake shear damage. Then the office should state the process gap plainly. The basement was not inspected, the omission was recorded, and nobody acted on the list. The final step is the change in practice. This order keeps the meeting focused on evidence first. It also turns acknowledgment of a process failure into a specific corrective action rather than a defensive argument.

TAKEAWAY 22W

Say what was found, what it means and what the process missed, in that order, and commit to the part you control.

10.4 The water table, while the trench is open

GEOTECHNICAL · A ROOM · HOLD

SCENE 50W

The trench for temporary shoring must stay open through the afternoon in saturated fill that liquefied nine days ago. Wellpoints keep the groundwater below the excavation. If the water rises too close to the floor, effective stress falls and seepage or local instability can develop before the shoring is complete.

GUIDE 43W

Keep the groundwater level inside the safe band by matching persistent inflows with pumping. The band narrows later in the exercise because the staged excavation leaves less hydraulic margin as work proceeds—not because sand automatically loses strength merely from being exposed to time.

ASK 13W

Hold the water table below the trench floor while the shoring goes in.

BEHIND THE BUTTON 89W

Why water level matters. Higher pore-water pressure reduces effective stress in the soil skeleton. Around an excavation, adverse hydraulic gradients can also drive seepage, piping or boiling.

Why inflow is a rate. Rain, a leaking main and harbour groundwater can continue feeding the excavation. A short burst of pumping does not balance a persistent inflow.

Why this is related to—but not the same as—liquefaction. Earthquake liquefaction is produced by cyclic loading that generates excess pore pressure. Excavation instability from groundwater is a static seepage/effective-stress problem unless new shaking occurs.

TAKES AS READ

water pressure between grains reduces the strength of sandy ground

VERDICT 81W

Raising groundwater increases pore pressure and reduces effective stress. In an open excavation, a sufficiently adverse hydraulic gradient can also produce seepage, piping or boiling. That is why the pump has to match continuing inflow rather than react only after the gauge has moved. This is not liquefaction 'arriving slowly': liquefaction requires cyclic loading to generate excess pore pressure. The game narrows the permitted band as the staged work proceeds to represent a smaller allowed hydraulic margin while shoring is incomplete.

TAKEAWAY 23W

Dewatering protects an excavation by controlling pore pressure and hydraulic gradients; that is related to effective stress but is not slow-motion earthquake liquefaction.

END OF THE DAY 21W

Evidence does not become weaker because it arrived outside the official process; it becomes useful when the process can test it.

DAY 11 OF 15

A quiet day on the bench

PLAN CARD 109W

Monday, and for the first time in ten days nothing is on fire. The Flats have their programme. The hospital is open. The school is running in its classrooms. Marisol Okonkwo spends a day like this on the half of town nobody is asking about. Upper Town was cleared fast, on the strength of still standing up. Standing up is not the same as undamaged. Alan Whitcombe walks the Parade with her this morning. The street is heavy old brick. You need to know what ties a wall like that to the floors behind it. Today you decide what is worth doing to buildings that are nobody's emergency yet.

BRIEFING 14W

Upper Town has been assessed, cleared and forgotten. Somebody should look at it properly.

WHAT YOU DECIDE 20W

Use the quiet day to understand why Upper Town survived and turn one repeated parapet failure into a targeted retrofit.

11.1 Walls that fall outward

STRUCTURAL ASSESSMENT · A ROOM · CHAIN

SCENE 40W

Owners are asking when the Parade can reopen. Every parapet fell outward into the street, while the buildings behind them remain standing. Timber floors are still connected. The fallen masonry is tagged by frontage, and the barriers are still up.

GUIDE 53W

Follow the sideways force from the shaking ground up through the building and name the link that gave way. Each step carries a reading saying what restraint it can actually provide. The heaviest element on the board is not automatically the one that failed, and neither is the first step in the path.

ASK 13W

Trace the sideways force through the frontage and name the link that failed.

BEHIND THE BUTTON 163W

Why out of plane. Masonry is strong in compression and weak in tension, so a wall pushed sideways cannot hold itself up by bending. It stays standing only while something ties it back at intervals — a floor, a roof, or a strap into the diaphragm. Where the tie stops, the wall is a free cantilever.

Why the parapet and not the wall. The wall below is held by the timber floors, which are still connected and did their job. The parapet projects above the roof line with nothing above it and nothing behind it, so the whole horizontal force it collects has to be resisted at its base, in bending, by mortar.

Why this is the useful kind of failure. It is local, repeatable and explained by one missing connection, so the retrofit targets the tie rather than the building. A frontage whose parapet is anchored back to the roof structure is a different proposition from one whose whole wall needs work.

TAKES AS READ

masonry is strong in compression and weak in tension

VERDICT 100W

Horizontal shaking gives the parapet an out-of-plane inertia force. Below roof level, the wall is restrained by connected floors and can transfer load into the diaphragms and cross walls. Above the roof, the parapet in this scenario lacks a positive brace or anchor back to the structure. The unsupported top therefore behaves much more like a cantilever, and weak masonry/mortar at the base can fail in bending or rocking. The useful diagnosis is the missing restraint in the load path; the retrofit must be designed around a verified anchorage capacity rather than assuming every surviving wall has the same detail.

TAKEAWAY 23W

Unbraced parapets are vulnerable to out-of-plane inertia; the missing restraint can make a local connection retrofit more important than strengthening the whole wall.

11.2 What you can learn without breaking it

MATERIALS & TESTING · A ROOM · VALUE

SCENE 37W

Ferreira has four test proposals for the Parade's listed brickwork. The scaffold permit allows one small trial location this week, and owners will not accept large-scale destructive sampling. A tie-back design still needs a defensible capacity number.

GUIDE 62W

Four tests are on the table and the permit allows one trial location. Open each card: it says what the test measures and how much listed fabric it consumes. The tie-back design needs a defensible pull-out capacity, so the test worth buying is the one that measures that property — not the one that produces the most precise number about something else.

ASK 10W

Which measurement answers the capacity question the retrofit depends on?

BEHIND THE BUTTON 164W

What the design actually needs. A tie-back is only as good as what it is anchored into, so the number the drawing needs is how much force an anchor in this brickwork can carry before it pulls out. Compressive strength, mortar composition and moisture content are all real properties of the wall, and none of them is that number.

Why precision is not value. A laboratory can give three significant figures on a property the design does not use, and the report will look better than a crude pull-out test with wide scatter. The question is which measurement changes the decision, which is the same question VALUE asks in every game that has one.

Why the fabric cost is on the card. This is listed brickwork: every test damages something that cannot be replaced, and the owners have already refused large-scale sampling. A test's cost here is not money but heritage, which makes choosing well a conservation decision as much as an engineering one.

TAKES AS READ

a test can be destructive, semi-destructive or neither

VERDICT 80W

The design question is whether an anchor installed in this old wall can transfer the required force. A trial-anchor pull-out test measures that property directly in the existing masonry and at the intended embedment. Brick compressive strength, rebound hardness and mortar condition can all describe parts of the wall, but none is the same as anchor pull-out capacity. The listed façade also makes disturbance a real cost. The best test therefore couples decision relevance with the smallest adequate physical intervention.

TAKEAWAY 18W

A precise test is valuable only when it measures the property that the pending design decision actually needs.

11.3 Before the next one

SEISMIC NETWORK · A PERSON · SCIENCETANK

SCENE 32W

Nothing on the Parade is dangerous today. The same parapets, on the same walls, will be here for the next earthquake, and the recurrence interval on this fault is about 140 years.

GUIDE 83W

Every parapet on the Parade came down and no wall behind one did. Tying back costs about £8,000 a building while the scaffolding stands, and about £25,000 once it is gone. A detailed assessment of one building takes three weeks and produces no strengthening. The grant scheme opens in eleven months and is oversubscribed in its first week. Lightweight replacement needs the parapets down first and a consented design, which is next summer at the earliest. The last event was six days ago.

ASK 6W

Decide what is worth doing now

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

a hazard that has occurred will occur again

VERDICT 84W

Tying parapets back has an unusually strong case: the failure mode occurred repeatedly, the retrofit acts directly on it, and scaffolding already in place makes the marginal cost much lower. The quoted 140-year recurrence interval is not a promise of 140 quiet years after this event; earthquakes do not reset a calendar. That statistic belongs in long-term hazard context, not in the argument for waiting. Act now because the defect is known, the consequence is falling masonry, and the cheapest access window is open.

TAKEAWAY 25W

> Fix a repeatedly observed, low-cost life-safety failure mode while access is already in place; do not let an average recurrence interval become a countdown clock.

END OF THE DAY 14W

A surviving building can reveal the load-path detail that prevented a much larger failure.

The record that was wrong all along

PLAN CARD 122W

Tuesday, and Inês Cardoso has found the reference error. The station above Upper Town was treated as competent bedrock, but the 1998 trench log shows several metres of weathered, fractured granite beneath it. A temporary sensor on nearby competent rock has now recorded five days of common aftershocks with the vault. In the 1 Hz band used for the Flats comparison, the vault motion is about 1.6 times the competent-rock motion. That means the published Flats-to-vault spectral ratio of about 3 was understating the Flats-to-rock ratio. The correction matters. So does a second mistake: the district planning memo copied that single-band ratio straight into a universal design-demand factor. You have to decide what gets corrected, what gets withdrawn and what remains untouched.

BRIEFING 18W

The reference station is not on competent bedrock. One correction now runs through every conclusion that used it.

WHAT YOU DECIDE 27W

Trace every conclusion that depended on the reference station, correct the measured site-response ratio and withdraw the design shortcut that copied one ratio into every future building.

12.1 Measuring against the wrong zero

SEISMIC NETWORK · A ROOM · TRACE

SCENE 40W

Cardoso's five-day side-by-side check is finished, and the benchmark used since day one no longer stands unchanged. Navarro lays five fortnight conclusions beside the corrected station report. Some may survive untouched; others may force a redraw before the next briefing.

GUIDE 60W

Open a conclusion and the panel shows what it was computed from. Keep the ones that stand on a measurement of their own, and untick the ones whose chain runs through the corrected reference. Then name the source the failing ones share. Both halves are graded: throwing away an independent conclusion costs you as much as keeping a dependent one.

ASK 14W

Which conclusions actually inherit the bad reference, and which remain supported by independent evidence?

BEHIND THE BUTTON 169W

What a shared reference is. Every station's record is interpreted against an assumed site response, and one station's has been corrected after five days of side-by-side comparison. Anything computed through that station inherits the correction; anything measured elsewhere does not. The error is not in the instrument — it is in one number every dependent conclusion was divided by.

Why untangling matters more than caution. The safe-looking move is to withdraw the whole fortnight, and it is wrong twice over: it throws away conclusions that were never affected, and it tells the next briefing that nothing is known when several things are. A retraction is a claim too, and it should be as narrow as the evidence.

How to read a chain. A conclusion is supported if some path to it never touches the suspect source. That is a question about the graph rather than about the size of the numbers, which is why the panel makes you open each one instead of judging by how alarming it looks.

TAKES AS READ

a comparison is only as good as the thing compared against

VERDICT 97W

Dependency tracing separates two issues. First, the measured Flats-to-vault ratio inherits the reference-station error: if the vault itself is about $\times 1.6$ relative to competent rock in the same band, a published $\times 3$ Flats-to-vault ratio becomes about $\times 4.8$ Flats-to-rock for that measured band. Second, the planning memo made an additional inference by using a single event- and frequency-specific ratio as a universal design multiplier. That shortcut should be withdrawn rather than merely changed from 3 to 4.8. The hospital record, multi-station M_w solution and Marina Court field diagnosis do not depend on the vault and therefore remain supported.

TAKEAWAY 25W

Trace both the measurement dependency and the inference built on top of it: the first may be corrected while the second must sometimes be withdrawn.

12.2 What still stands

GEOTECHNICAL · A ROOM · CASEBOOK

SCENE 22W

Navarro puts four of the fortnight's conclusions on the board. One used the suspect vault directly; the others have independent evidence chains.

GUIDE 39W

Trace each conclusion back to its evidence. The 1 Hz Flats spectral ratio used the vault as its denominator. The hospital's period-specific record did not. Marina Court's rigid-body rotation and ejecta did not. The Parade's parapet failures did not.

ASK 4W

Sort the fortnight's conclusions

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a conclusion can rest on one measurement or on several

VERDICT 75W

Trace each conclusion to its evidence. The Flats spectral ratio used the vault directly and must be restated. The district design shortcut also depended on that ratio, but it should be withdrawn rather than assigned a new universal factor because site response varies with period and input motion. Marina Court's diagnosis came from survey, ejecta and geotechnical observations. The hospital motion came from its own instrument, and moment magnitude came from the network source solution.

TAKEAWAY 16W

A reference error propagates through dependent measurements, not through every conclusion made during the same emergency.

12.3 Four point eight — in one band

HAZARD & FORECASTING · INCIDENT BASE · A PERSON · BALLPARK

SCENE 44W

The dependency trace leaves one clean numerical correction. In the 1 Hz band used in the original comparison, the Flats-to-vault spectral ratio was about 3.0. Five days of common aftershocks show the vault itself is about 1.6 times competent rock in that same band.

GUIDE 45W

Multiply ratios only because they refer to the same response quantity and frequency band: Flats/vault × vault/rock = Flats/rock. Then stop. The result is a corrected 1 Hz spectral ratio, not 'the amplification of the Flats' at every period and not a replacement code coefficient.

ASK 13W

What is the corrected Flats-to-competent-rock spectral ratio in the measured 1 Hz band?

BEHIND THE BUTTON 89W

What is being corrected. Ratios can be chained when numerator and denominator describe the same quantity: (Flats/vault) × (vault/rock) = Flats/rock.

What the number means. About 4.8 is the corrected spectral-amplitude ratio in the specified 1 Hz band for the observed motions. Site response varies with frequency and can also depend on shaking level.

What the number cannot become. A building design spectrum contains period-dependent ordinates and code factors. One corrected spectral ratio is evidence for stronger site effects, not a universal multiplier to paste into every design value.

TAKES AS READ

both ratios refer to the same spectral quantity and the same 1 Hz band · peak ground acceleration and what a building feels — taken as read

VERDICT 69W

The measured chain is (Flats/vault) × (vault/competent rock) = 3.0 × 1.6 ≈ 4.8. Because both terms are dimensionless spectral-amplitude ratios in the same band, the multiplication is legitimate. What is not legitimate is the next leap: treating 4.8 as the response of every structure at every period or as a code design coefficient. The corrected result strengthens the case for a proper site-response study and period-dependent design spectrum.

TAKEAWAY 20W

A corrected spectral ratio can be scientifically valid while the policy shortcut built from the old ratio is still invalid.

12.4 What the teams are being sent to

PUBLIC SAFETY · A ROOM · SPOT

SCENE 38W

Buildings queue for assessment faster than the teams can reach them. The office works to one priority, and Okonkwo changes it as the fortnight does — after the aftershock, after the schools ask, after the cordon goes up.

GUIDE 53W

Send teams to what the current priority wants and leave the rest queued. The priority on the office board changes during the day and nobody announces it. What is scored is the buildings either side of a change, because they are the only ones that show whether the office is reading the board.

ASK 12W

Dispatch to the priority on the board, and keep watching the board.

BEHIND THE BUTTON 92W

Why the priority changes. Early it is occupied buildings, because people are in them. After an aftershock it is anything already placarded yellow, because their capacity has fallen and the same shake now does more. When the schools ask it is schools, because a decision is waiting on the answer rather than on the risk.

Why the changeover costs. A team sent under the old priority spends half a day on a building nobody is waiting for, and the building that the new priority named waits another night with people in it.

TAKES AS READ

an office decides which buildings are assessed first

VERDICT 116W

Three priorities run across the fortnight and each wants a different part of the queue. Occupied buildings, then anything already placarded yellow, then schools. A building can answer two at once, which is what makes the change cost real rather than notional, and the panel scores the window either side of each change because most of the queue is wanted by neither priority and is correctly left waiting by somebody who has read nothing. The office's own version is the hour after an aftershock: the priority has changed to the yellow placards, whose capacity has just fallen again, and the teams already in the vans are driving to the list that was current before the shake.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

END OF THE DAY 26W

A bad reference contaminates only the conclusions that depend on it; correcting a measurement does not justify turning one event-specific ratio into a universal design factor.

Three things at once

BEFORE THE DAY — HUNT

Six props went out and the log has four

Every temporary prop on the Parade has to be accounted for before the cordon can be reduced, because a prop that nobody can find is either holding something up or lying in a street where a lorry will hit it. Four are where the log says. Find the rest.

PLAN CARD 102W

Wednesday, 15:40, and 3 things arrive together. A magnitude 5.1 aftershock has just shaken the town. It is the largest in a week. A water main has burst on Ferry Street. It is pouring into ground everybody now knows is loose. And the district wants a call on the school gym by five o'clock. Contractors are holding a slot for Saturday. Marisol Okonkwo has four engineers and no more. Bram Halvorsen is on the phone to the water board. Funmi Adeyemi has three drafts of a notice nobody has signed off. Today you decide which of the three gets your engineers first.

BRIEFING 15W

An aftershock, a burst main and a deadline for the school, all inside an hour.

WHAT YOU DECIDE 22W

Triage an aftershock, a burst main and a school deadline by asking which hazard is changing fastest and which decision closes first.

13.1 Which of the three cannot wait

PUBLIC SAFETY · A ROOM · DELEGATE

SCENE 36W

Three calls land within minutes. A burst main is feeding water into the Flats. Several vulnerable buildings need checks after the M5.1 aftershock, and Bay Road School needs its reopening message issued. Four engineers are available.

GUIDE 66W

Three calls, four engineers, and one of the three you have to keep yourself. For each of the others, choose an owner who can take a real first action and the reading or event that brings the problem back to you. A handover with an owner and no return condition is not a handover. Then take the watch on the one that needs your own judgement.

ASK 27W

How do you divide the team so the worsening condition is stopped, the life-safety consequence is checked and the stable deadline does not consume the field response?

BEHIND THE BUTTON 167W

Which one to keep. One of these is getting worse while you decide, one has a life-safety consequence that has to be checked, and one is a deadline that is stable — the message will still need issuing in an hour. Keeping the stable one is the classic error: it is the easiest to finish, and finishing it costs the field response its hours.

Why a first action is required. "Watch it" is not a handover. An owner who cannot take a concrete step is a person who will come back for instructions at the worst moment, so the panel asks for the action they will take before they need to ask anybody.

Why a return condition is the other half. Delegation is not disposal: the problem comes back if it crosses a threshold, and naming that threshold now is what lets somebody else hold it without checking in. A burst main over liquefiable fill and a reopening message have very different triggers, and both need one.

TAKES AS READ

a problem still growing is different from one that has finished

VERDICT 76W

The burst main is actively worsening the ground condition, so utility isolation starts immediately. The M5.1 aftershock is over, but vulnerable or shored buildings may have new distress. That uncertainty needs engineering judgment before restrictions change. The school decision is already technically settled and mainly needs communication. Parallel work is therefore better than a single ranked list. Stop the growing input, keep direct attention on uncertain building consequences, and hand the stable message to the desk.

TAKEAWAY 28W

> Good delegation couples every handoff to a first action and a condition for returning the problem, while scarce expert attention stays on the judgment that cannot be automated.

13.2 Wetting ground that has already failed

GEOTECHNICAL · A PERSON · BALLPARK

SCENE 33W

Mbeki is on Ferry Street with a standpipe. The water table under the road has risen three quarters of a metre since the main went, and the ground surface is soft again underfoot.

GUIDE 70W

Put a number on what the leak did. The grains are held together by the effective stress, which is the total weight above them less the water pressure between them. Pick the total stress at three metres and the pore pressure the risen water table now gives, and take one from the other. Two tiles are the pressures before the main burst, which is the comparison rather than the answer.

ASK 18W

At three metres down under Ferry Street, what is the effective stress now the water table has risen?

BEHIND THE BUTTON 148W

What effective stress is. The soil skeleton carries the difference between the total stress pressing down and the water pressure in the pores pushing the grains apart. Strength lives in that difference, which is why a saturated fill can be strong one day and behave like a liquid the next without a gram of it moving away.

What the leak did and did not do. A higher water table raises the pore pressure and takes the margin down. It liquefies nothing on its own — cyclic shaking is still required to build the pressure the rest of the way. What has changed is how much shaking that now takes.

Why Upper Town is different. On competent granite the same burst main is a plumbing problem. There is no loose saturated skeleton whose grain contacts can be unloaded by water pressure, so the mechanism has nothing to act on.

TAKES AS READ

liquefaction requires saturated ground

VERDICT 99W

At 3 m depth, total vertical stress is about 18 kN/m^3 × 3 m = 54 kPa. With the water table now 1.5 m below ground, the pore pressure at 3 m is about 9.81 × 1.5 ≈ 14.7 kPa, so $\sigma' \approx 54 - 14.7 \approx 39$ kPa. Before the rise, $u \approx 7.4$ kPa and $\sigma' \approx 47$ kPa. The soil skeleton therefore carries less effective stress. That changes liquefaction susceptibility and should trigger a renewed geotechnical check, but cyclic resistance and demand do not scale one-for-one with this 16% change. The leak has not liquefied the ground by itself; renewed cyclic shaking would still be required.

TAKEAWAY 25W

A higher water table lowers effective stress and can worsen liquefaction susceptibility, but the triggering margin is not a simple linear percentage of effective stress.

13.3 A smaller shake against a weaker building

STRUCTURAL ASSESSMENT · A ROOM · CHOICE

SCENE 30W

The 5.1 was about a tenth of the mainshock's ground motion in the Flats. 11 buildings have been shored since Friday, and 2 of them have new cracking this afternoon.

GUIDE 54W

Four readings of the same afternoon. Ask of each what the shoring was for. It is temporary support, not a repair, so new cracking is not automatically a failure of it. What has changed since Friday is on the other side of the comparison. Eleven buildings have less capacity, and two of them responded.

ASK 11W

What do the two buildings with new cracking tell the office?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a shored building is being held rather than repaired · p and S waves, and what the gap between them measures — taken as read

VERDICT 62W

New cracking after a smaller aftershock is evidence of additional distress in already damaged structures. It justifies maintaining or tightening restrictions and checking the damaged mechanism and the shoring. It does not by itself locate the building on a precise capacity curve, prove the shoring failed, or invalidate the first assessment. The observed change is the reason to re-enter the evidence loop.

TAKEAWAY 23W

New cracking during a smaller aftershock is evidence that residual capacity or detailing deserves escalation; it is not a complete diagnosis by itself.

END OF THE DAY 15W

Priority comes from consequence, rate of change and reversibility—not from which problem sounds most dramatic.

DAY 14 OF 15

The last morning it can be changed

PLAN CARD 100W

Thursday, and the recovery plan goes to the council tonight. One clause in it matters more than the rest. It decides whether rebuilding in the Flats needs the ground itself improved, or only stronger foundations. Elena Navarro has the corrected amplification and the borehole programme. Bram Halvorsen has the cost. It is large, and it falls on people who have just lost houses. Yvette Delacroix has 400 households who want to rebuild this year. They cannot afford to wait for a scheme. After tonight the clause cannot be reopened for five years. Today you decide what goes into that clause.

BRIEFING 16W

The recovery plan closes for comment today, and the ground improvement clause is in or out.

WHAT YOU DECIDE 21W

Write the rebuilding rule so it reduces the Flats liquefaction hazard without pretending one treatment or one CPT number guarantees safety.

14.1 Fixing the ground instead of the building

GEOTECHNICAL · A ROOM · CHOICE

SCENE 33W

Navarro has three options costed: stone columns, deep densification, and doing nothing to the ground while designing stiffer foundations on top of it. The council wants one approach written into the rebuild guidance.

GUIDE 69W

Three costed options and four readings of what separates them. Ask of each option where in the system it acts: on the soil, on the structure, or on neither. A stiff foundation can carry a building across ground that moves. It does nothing for the road, the buried services or the site next door. Both approaches can be right, and the council is writing one of them into guidance.

ASK 16W

What is the engineering difference between improving the ground and designing foundations to tolerate the hazard?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

liquefaction can be addressed in the ground or accommodated in the structure

VERDICT 81W

The strategies act at different places. Densification can increase density and cyclic resistance; drains or stone columns may reduce excess pore-pressure buildup or its consequences, depending on the design. None of those guarantees that the soil can never liquefy. Piles, rafts or other foundation strategies can bypass weak layers or tolerate some deformation, but roads, utilities and neighbouring sites may still move. A town-level rule therefore has to state which consequence it is trying to reduce and what performance is required.

TAKEAWAY 18W

Ground improvement reduces liquefaction susceptibility or its consequences; structural foundations can instead bypass or accommodate some ground deformation.

14.2 Six minutes outside a building

STRUCTURAL ASSESSMENT · A PERSON · CASEBOOK

SCENE 33W

Whitcombe's form has 11 boxes and a place for the colour. He filled 400 of them in 2 days, and he has kept a second list of everything he did not look at.

GUIDE 71W

The left column is four things about a building. The right column says whether six minutes outside it could have reached them. Ask of each: could somebody see this from the footpath? A form with a colour on it will be read later as a claim about the whole building. So what was not looked at has to be recorded as carefully as what was, and Whitcombe kept that second list.

ASK 7W

Say what a rapid assessment actually samples

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an inspection covers what the inspector could see

VERDICT 82W

A rapid assessment is designed to screen many buildings quickly after a disaster. It can identify gross leaning, collapse hazards, obvious structural distress and accessible life-safety problems. It cannot establish the condition of concealed connections, inaccessible rooms, foundations or every member in the load path. The discipline is therefore two-part: record what was observed and record what could not be observed. A green placard is useful only when later users do not silently expand its scope beyond the inspection that produced it.

TAKEAWAY 17W

> A rapid assessment is triage: it records observed hazards and access limits, not a complete structural evaluation.

14.3 What the next five years hold

PUBLIC SAFETY · A ROOM · PROTOCOL

SCENE 35W

The clause goes in or it does not, and nobody can reopen it for five years. Delacroix is holding the spring rebuild schedule beside the draft ordinance. The geotechnical section is the last unresolved page.

GUIDE 63W

Four versions of the clause on the left, and what each one actually produces on the right. Pair them by asking who has to pay and how soon. A requirement nobody can afford stops rebuilding instead of improving ground. No requirement rebuilds the same vulnerability on the same fill. And the last pairing is about time: nobody can reopen this for five years.

ASK 7W

Commit, and say what the commitment contains

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a decision to require something is also a decision about who pays for it

VERDICT 85W

The engineering evidence can define the ground hazard and the performance a treatment programme should demonstrate. It cannot decide who pays or how quickly households must comply. A requirement without financing can stall rebuilding. No requirement can reproduce the same vulnerability. A workable policy therefore states where the rule applies and names a measurable geotechnical acceptance target. It also provides funding or phasing and a review process as new site data arrive. That separates the technical standard from the political choices needed to implement it.

> **TAKEAWAY** 21W
>
> Science can define the hazard and performance target; the rebuilding policy must also specify affordability, phasing and who carries the cost.

END OF THE DAY 19W

A rebuilding standard should state the hazard it addresses, a measurable acceptance criterion and the limits of that criterion.

What we know, and how well

PLAN CARD 113W

Friday, two weeks on. The fortnight now goes into a report. Marisol Okonkwo will not sign a report that states everything at the same strength. Some of it was measured. Some of it was judged. A reader who cannot tell those apart will act on both the same way. Elena Navarro has the borehole results. Rei Tanaka wants her forecast written as a rate with a range. Not as a single number. Inês Cardoso wants the vault error set down as a finding. Not buried as a fix. Funmi Adeyemi has to turn all of it into something a household can use. Today you sort the findings by the evidence under each one.

BRIEFING 27W

The Placard Register goes out today. The last job is not another calculation; it is deciding whether the evidence under every door is strong enough to sign.

WHAT YOU DECIDE 22W

Grade each finding by its evidence, state the remaining uncertainty and sign only the Placard Register that says what is still unresolved.

15.1 Measured, inferred, assumed

SEISMIC NETWORK · A ROOM · CASEBOOK

SCENE 32W

Cardoso lays out four of the fortnight's statements. The office believes all four. They are not held for the same reasons and she will not let them into the report as equals.

GUIDE 63W

Four statements the office believes, held for four different reasons. Sort them by what an opponent would have to produce to challenge each: a second log, another event, a borehole, or a future earthquake. Belief is not the variable. The weakest of the four is the one the plan clause rests on, and saying so in the report is what makes it usable.

ASK 6W

Sort the findings by their evidence

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a claim can rest on a measurement, on an inference, or on nothing yet

VERDICT 89W

The findings deserve different labels because their evidence chains differ. The vault/reference correction is directly documented by the trench log and common-event comparison. The hospital's period-specific recorded motion is measured at one building for one event. Marina Court's mechanism is a strong inference from rigid-body rotation, ejecta and ground observations without direct access beneath the raft. Future Flats site response is a projection that requires a period-dependent site-response study; the corrected 1 Hz ratio is not a universal future multiplier. The report becomes stronger when those distinctions remain visible.

TAKEAWAY 17W

> Sorting findings by their evidence is what lets a later reader know which ones to check first.

15.2 The number that has to be a range

HAZARD & FORECASTING · INCIDENT BASE · A PERSON · CHOICE

SCENE 35W

Tanaka's aftershock forecast goes into the report. She wants the range in the sentence and Okonkwo wants to know what the range is for. The staffing appendix cannot be signed until she chooses the wording.

GUIDE 68W

Four reasons to publish a range, and they differ in who the range is for. Ask of each whether it is about caution, honesty, defensibility, or planning. Nobody staffs an emergency to the average week. A single number tells a planner nothing about how far the sequence might run above it, so they invent a margin instead. The staffing appendix cannot be signed until the wording is chosen.

ASK 17W

Why must the aftershock forecast include a range or probability distribution rather than only one expected count?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a forecast is a distribution rather than a single value

VERDICT 61W

An aftershock forecast is probabilistic. The expected count alone hides the spread of plausible outcomes, while staffing and inspection capacity are decisions made against that spread. Publishing the distribution or a clearly defined interval also makes the forecast testable and keeps model uncertainty visible. The range is not decorative caution and it does not mean the model lacks an expected value.

TAKEAWAY 17W

> A forecast without a range cannot size a decision, because the decision is sized against the range.

15.3 Sign the register

STRUCTURAL ASSESSMENT · A ROOM · TRIAGE

SCENE 44W

The fifteen entries are on the table. Okonkwo has left one blank line above the signature. It is for the condition under which you are willing to sign: what happens to every uninspected space and every unresolved hazard after today's report leaves the room.

GUIDE 61W

Choose the register you can defend. A signed record may contain uncertainty, but it cannot hide it. The school gym, hospital plant room and Ferry Street basement all began as written inspection limits that nobody owned. The final version must turn every unresolved item into an assigned action with a deadline or trigger before a later full clearance can erase it.

ASK 9W

Which version of the Placard Register can you sign?

BEHIND THE BUTTON 88W

Why this is the final decision. Every earlier placard was provisional evidence under pressure. The register is the durable record other people will use after the emergency team disperses.

What makes uncertainty acceptable. An unresolved condition can remain in a signed report if its consequence, current restriction, owner and next check are explicit. Hidden uncertainty is the failure.

What the fortnight revealed. The repeated blind spot was not that inspectors failed to see through walls; it was that written limits of inspection were allowed to become nobody's task.

TAKES AS READ

a safety coordinator may sign a report that contains explicit unresolved items and restrictions

VERDICT 80W

The defensible register does not claim that every building is safe. It states the current use restriction, the evidence behind it, the parts not inspected, and who owns each unresolved item with a deadline or trigger for reassessment. That is the lesson shared by the school gym, hospital plant room, Ferry Street basement and reference-station error: uncertainty becomes dangerous when the record makes it disappear. Signing the qualified register is the actual decision the fifteen days have been building toward.

TAKEAWAY 18W

Sign only the register that preserves unresolved hazards as owned actions rather than silently converting them into clearances.

END OF THE DAY 16W

The final safety decision is defensible only when measurements, inferences and unresolved unknowns are clearly separated.

THE CLOSING CARD

At four on Friday afternoon, you sign the qualified register. The last line does not say every building is safe. It says which spaces are still closed, which unknowns have owners, and what evidence is required before the next placard changes. Families begin returning street by street. The hospital stays open. Bay Road School keeps its classrooms and closes the gym until its panel connections are repaired. The Parade stays behind barriers until its parapets are positively tied back.

The biggest correction is not a colour on a door. The station everybody treated as competent-rock reference was sitting over weathered ground. The Flats site-response ratio is stronger than the first report said in the measured band, and the district withdraws the shortcut that had turned one spectral ratio into a universal design multiplier. The rebuilding rule now requires a real site-response and geotechnical check instead. Marina Court remains red and is scheduled for demolition after its rigid-body tilt is documented and the neighbouring ground is investigated.

Weeks later the cordons come down only when the hazard written against each one has been dealt with. What changed Kestrel Bay was not that you found a perfect test. You kept asking what each inspection had actually seen, what each measurement was really measuring, and what remained unknown. Four hundred households got decisions they could act on, and the next coordinator inherits a register that says exactly how far every one of those decisions can be trusted.

Card text only — no options, boards or answers. 11,299 words of card prose across 52 stops. Generated from `themes/aftershock` by `tools/make-cardbook.mjs`.